Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT **Paper -2(Adv-2024-P2-Model)** **Date: 21-12-2025**

Time: 02.00Pm to 05.00Pm **GTA-08** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**B**	2	**C**	3	**A**	4	**A**	5	**BD**	6	**BD**
7	**BCD**	8	**743**	9	**22**	10	**7**	11	**2**	12	**8**
13	**25**	14	**1.75**	15	**0.86 – 0.87**	16	**4.1**	17	**1.57**		

PHYSICS

18	**C**	19	**A**	20	**A**	21	**C**	22	**BCD**	23	**AD**
24	**ABC**	25	**5**	26	**8**	27	**9**	28	**4**	29	**0**
30	**3**	31	**2**	32	**3**	33	**2**	34	**30**		

CHEMISTRY

35	**C**	36	**C**	37	**D**	38	**B**	39	**AB**	40	**ABC**
41	**D**	42	**9**	43	**4**	44	**4**	45	**0**	46	**3**
47	**15**	48	**6**	49	**63**	50	**82**	51	**-63**		

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHEMATICS

1. $2x-y+1=0$

$3x-y=0$

$2x+y-5=0$

$\Rightarrow\ x=1, y=3$

$\therefore\ \text{Least distance}=\left|\dfrac{3-12+19}{5}\right|=2$

2. Let $x+y+z=\theta \text{ and } k=2$

$\therefore\ \cos x+\cos y+\cos z=k\cos\theta \text{ and } \sin x+\sin y+\sin z=k\sin\theta$

$\therefore\ \cos(x+y)+\cos(y+z)+\cos(z+x)=\cos(\theta-z)+\cos(\theta-x)+\cos(\theta-y)$

$=\cos\theta(k\cos\theta)+\sin\theta(k\sin\theta)=k=2$

3. $I=\int_{-20}^{20}\dfrac{f(x)}{g(x)}dx$

Using King's rule, we have

$I=\int_{-20}^{20}\dfrac{f(x)}{g(x)}dx=\int_{-20}^{20}\dfrac{f(-x).f(x)}{g(x)}dx$

Addition the two, we get

$:2I=\int_{-20}^{20}\dfrac{f(x).(1+f(-x))}{g(x)}dx=\int_{-20}^{20}\dfrac{f(-x).(1+f(-x))}{g(x)}dx=\int_{-20}^{20}f(-x)dx=20$

$\Rightarrow I=1010$

4. Draw the graph we get $P_2\in(1.5,7)$

5. Shift LHS integral to RHS to get

$\int_{2}^{150}\left(f^2(x)-(x-1)\ln(x-1)\left(2f(x)-(x-1)\ln(x-1)\right)\right)dx=0$

$\Rightarrow\ \int_{2}^{150}\left(f^2(x)-2f(x)\ln(x-1)+\left((x-1)\ln(x-1)\right)^2\right)dx=0$

$\Rightarrow\ \int_{2}^{150}\left(f(x)-(x-1)+\ln(x-1)^2\right)dx=0$

But, $\left(f(x)-(x-1)+\ln(x-1)\right)^2\geq 0$

So, $\left(f(x)-(x-1)+\ln(x-1)\right)^2=0$

$\Rightarrow\ f(x)-(x-1)+\ln(x-1)=0$

$\Rightarrow\ f(x)=(x-1)+\ln(x-1)$

Option a is incorrect because area is equal to $\dfrac{1}{4}$.

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6. $f(x)=(x-2)^2+6$

7. $\begin{bmatrix}a_{n+1}\\b_{n+1}\end{bmatrix}=\begin{bmatrix}\sqrt{3} & -1\\1 & \sqrt{3}\end{bmatrix}\begin{bmatrix}a_n\\b_n\end{bmatrix}=2\begin{bmatrix}\cos\frac{\pi}{6} & -\sin\frac{\pi}{6}\\\sin\frac{\pi}{6} & \cos\frac{\pi}{6}\end{bmatrix}\begin{bmatrix}a_n\\b_n\end{bmatrix}$

$\therefore \begin{bmatrix}a_2\\b_2\end{bmatrix}=2\begin{bmatrix}\cos\frac{\pi}{6} & -\sin\frac{\pi}{6}\\\sin\frac{\pi}{6} & \cos\frac{\pi}{6}\end{bmatrix}\begin{bmatrix}a_1\\b_1\end{bmatrix}$

$\therefore \begin{bmatrix}a_3\\b_3\end{bmatrix}=2^2\begin{bmatrix}\cos\frac{2\pi}{6} & -\sin\frac{2\pi}{6}\\\sin\frac{2\pi}{6} & \cos\frac{2\pi}{6}\end{bmatrix}\begin{bmatrix}a_1\\b_1\end{bmatrix}$

$\therefore \begin{bmatrix}a_{100}\\b_{100}\end{bmatrix}=2^{99}\begin{bmatrix}\cos\frac{99\pi}{6} & -\sin\frac{99\pi}{6}\\\sin\frac{99\pi}{6} & \cos\frac{99\pi}{6}\end{bmatrix}\begin{bmatrix}a_1\\b_1\end{bmatrix}$

$\Rightarrow a_1=\frac{1}{2^{97}};b_1=-\frac{1}{2^{98}}$

8. $(EM)^T=20I$

Take transpose on both sides

$EM=20I$.....(1)

$(E+M)^T=17(E-M)^T$

$E^T+M^T=17(E-M)^T$

$16E^T=18M^T$

Take transpose on both sides

$16E=18M$ .....(2)

From Eqns (1) and (2), we get

$E=\pm\frac{3\sqrt{20}}{2}I;M=\pm\frac{4\sqrt{20}}{3}I$

$E^2+M^2=\frac{725}{18}I\Rightarrow a+b=743$

9. Let $P(x)=\sum_{k=1}^{n}kx^k\equiv n(x-a_1)(x-a_2).....(x-a_n)$

$\ln P(x)=\ln+\sum_{i=1}^{n}\ln(x-a_i)$

Differentiate two times and put $x=1$, we get

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$$\frac{P(1)P''(1)-(P'(1))^2}{(P(1))^2}=\sum_{k=1}^{n}\frac{1}{(1-a_k)^2}=13$$

Now, $P(x)=\sum_{k=1}^{n}kx^k$

$\Rightarrow P(1)=\sum_{k=1}^{n}k=\frac{n(n+1)}{2}$

$P'(1)=\sum_{k=1}^{n}k^2=\frac{n(n+1)(2n+1)}{6}$

$P''(1)=\sum_{k=1}^{n}k^2(k-1)=\sum n^3-\sum n^2=\left(\frac{n(n+1)}{2}\right)^2-\frac{n(n+1)(2n+1)}{6}$

On putting these values we get $n=22$.

10. $S_k=\int_0^1 x^2(1-x)^k\,dx=\int_0^1(1-x)^2x^k\,dx$ $\left(use=\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right)$

Hence $\sum_{k=1}^{\infty}\int_0^1(1-x)^2x^k\,dx=\int_0^1(1-x)^2\underbrace{\sum_{k=1}^{\infty}x^k\,dx}_{\text{infinite}\,G.P.}$

$=\int_0^1(1-x)^2\left(\frac{x}{1-x}\right)dx=\int_0^1(1-x)^2dx=\frac{1}{2}-\frac{1}{3}=\frac{1}{6}\Rightarrow p+q=7$

11. $P_1: x+y+z=1$

$P_2: x+y+z=\frac{9}{2}$

$P_3: 2x-5y+z=-5$

Vector along the line of intersection of the planes $\frac{P_1}{P_3}$ or $\frac{P_2}{P_3}$ is

$$=\vec{n}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&1\\2&-5&1\end{vmatrix}=6\hat{i}+\hat{j}-7\hat{k}$$

Point on line L_1 which is the line of intersection of P_1 and P_3 say $(x_1,y_1,0)$

Hence, $x_1+y_1=1$ and $2x_1-5y_1=-5$

Solving, we get $y_1=1$ and $x_1=0$

Hence, point on L_1 is $(0,1,0).....(A)$

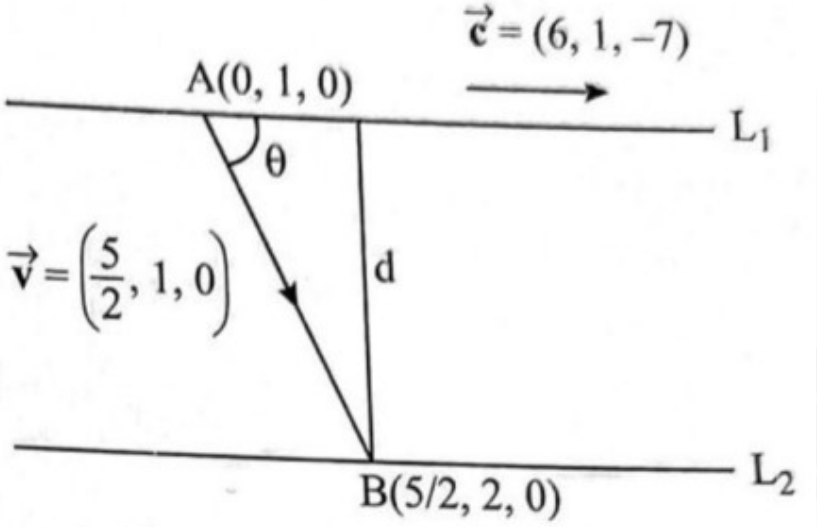
$||ly$ point on line L_2 which is line of intersection of P_2 and P_3 say $(x_2,y_2,0)$

Hence $x_2+y_2=\frac{9}{2}$ and $2x_2-5y_2=-5$

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Solving, we get $y_2 = 2$ and $x_2 = \frac{5}{2}$

Hence, point on L_2 is $\left(\frac{5}{2}, 2, 0\right)$.....$(B)$



Now $d = |\vec{v}|\sin\theta = \frac{|\vec{v}\times\vec{c}|}{|\vec{c}|} = \frac{\sqrt{|\vec{v}|^2|\vec{c}|^2 - (\vec{v}.\vec{c})^2}}{|\vec{c}|}$.....$(1)$

Now $|\vec{v}| = \sqrt{\frac{25}{4}+1} = \sqrt{\frac{29}{4}}; |\vec{c}| = \sqrt{86}$

$\vec{v}.\vec{c} = 16$

Now from equation (1), $d = \sqrt{\frac{\frac{29}{4}\times 86 - 256}{86}}$

$= \sqrt{\frac{735}{2\times 86}} = \sqrt{4.27} = 2.06$

12. $\pi\left(f'(x)\right)^2\cos\left(\pi\left(f(x)\right)\right) + \sin\left(\pi\left(f(x)\right)\right)f''(x) = \frac{d}{dx}\left(\sin\left(\pi f(x)\right)\times f'(x)\right)$

Let $g(x) = \sin\left(\pi f(x)\right)\times f'(x) in [\alpha,\beta]$

$f'(x) = 0$ has one root in $[\alpha,\beta]$

Now, $\sin\left(\pi f(x)\right) = 0 \Rightarrow \pi f(x) = k\pi (k \in I) \Rightarrow f(x) = k$

Now, $f(x) = 3,2,1,0$ for each value we will get 2 values of x.

Hence minimum number of roots of $g(x)$ in $[\alpha,\beta]$ is 9.

Using Rolle's Theorem minimum number of $g'(x)$ is 8.

13. $a = 5, \cos B = \frac{3}{7}, c = 7$

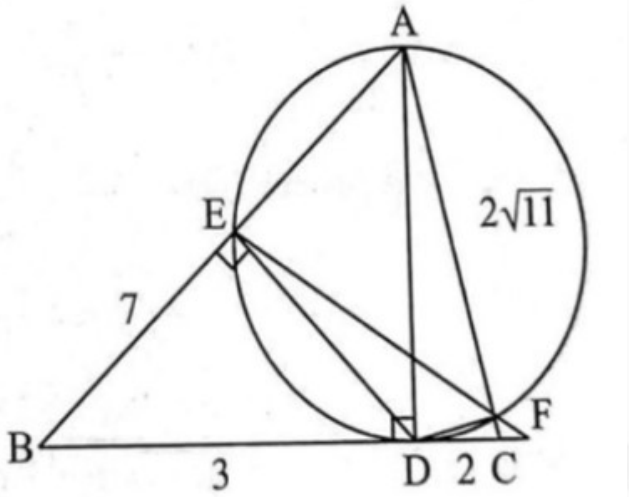
$BE = 3\cos B = \frac{9}{7}$

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$$CF = 2\cos C = 2\times\frac{1}{\sqrt{11}}$$

$$\therefore\ (BE)(CF) = \frac{18}{7\sqrt{11}} \equiv \frac{p}{q\sqrt{11}}$$

$$\therefore\ p+q = 25$$



14, 15

$$P = 12x + 4K$$

$$A = 6x^2 + 7Kx$$

$$p = 30 \Rightarrow A = 6x^2 + \frac{7x(P-12x)}{4}$$

$$A = 6x^2 + \frac{7xP}{4} - 21x^2$$

$$= \frac{7Px}{4} - 15x^2$$

$$\text{A is minimum} \Rightarrow 30x = \frac{7P}{4} \Rightarrow 30x = \frac{7}{4}\times 36 \Rightarrow x = \frac{7}{4} = 1.75$$

$$A = 6x^3 + 7Kx = 13 \Rightarrow K = \frac{13-6x^2}{7x}$$

$$P = 12x + 4Kx = 12 \Rightarrow + \frac{4\left(13-6x^2\right)}{7x}$$

$$P = 12x + \frac{52}{7x} - \frac{24x}{7}$$

$$= \frac{60x}{7} + \frac{52}{7x} \text{is minimum}$$

$$\text{When } 60x = \frac{52}{7} \Rightarrow x^2 = \frac{52}{60} = \frac{13}{15}$$

16, 17.

$$f(x) = ax^2 + bx + c$$

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$g(x)=a\ln^2+b\ln x+c\ \forall x>0$

Given, $g'(p)=0$

$\Rightarrow \dfrac{2a\ln p+b}{p}=0 \Rightarrow \ln p=\dfrac{-b}{2a}$

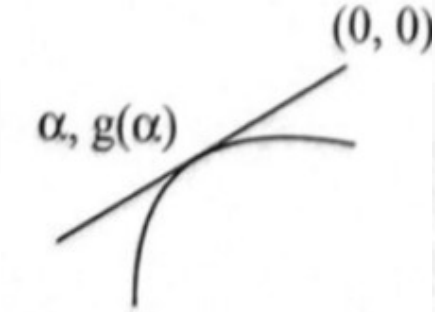
Also $g''(p)^2=0 \Rightarrow \dfrac{2a\left(1-\ln p^2\right)}{p^4}=0$

$\Rightarrow 1-\dfrac{b}{2a}=\ln p^2 \Rightarrow \int_1^2 \dfrac{2a}{2\left[(n-1)^2+1\right]}$

$\therefore\ 1-\dfrac{b}{2a}=2\ln p=\dfrac{-b}{a} \Rightarrow 2.\left(Tan^{-1}(x-1)\right)_1^2$

$\therefore\ b=-2a$

Also,



$\dfrac{100-20+2}{4-4+2}=\dfrac{82}{4}=41$

$2x-2$

$=2\left[\dfrac{\pi}{4}-0\right]=\dfrac{\pi}{2}=1.57$

$g'(\alpha)=\dfrac{g(\alpha)-0}{\alpha-0}$

$\dfrac{2a\ln\alpha+b}{\alpha}=\dfrac{a\ln^2\alpha+b\ln\alpha+c}{\alpha}$

$a\ln^2\alpha+(b-2a)\ln\alpha+c-b=0$

For unique $\alpha, D=0$

$(b-2a)^2-4a(c-b)=0$

$4b^2=4a(c-b)$

$4a=c+2a \qquad \therefore\ c=2a$

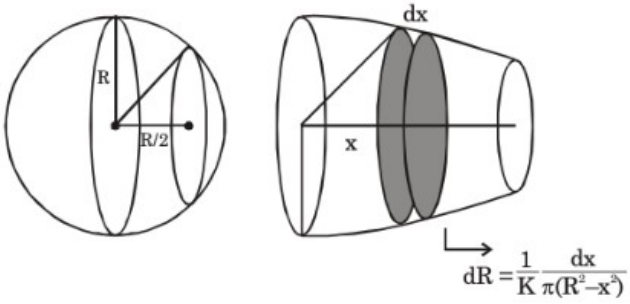
$\therefore f(x)=a\left(x^2-2x+2\right)$

Now, proceed.

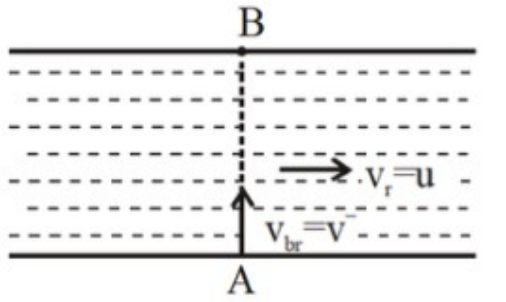
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PHYSICS

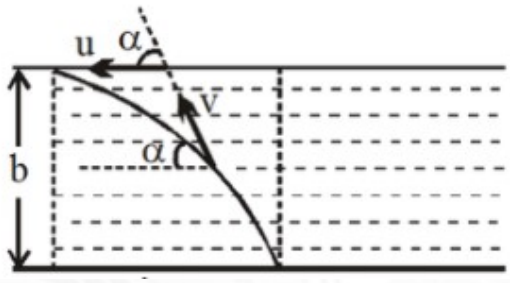
18. $\int \frac{1}{K}.\frac{dx}{\pi\left(R^2-x^2\right)}$ $R_T=\frac{1}{K\pi}\int\frac{dx}{\left(R^2-x^2\right)}=\frac{1}{2K\pi R}\ln(3)$



19.



In the river frame dock will move towards left with speed u



$$ut_0=v\int_0^{t_0}\cos\alpha\,dt \quad .....(1)$$

$$v-u\cos\alpha=-\frac{ds_{rel}}{st}$$

$$\int_0^{t_0}(v-u\cos\alpha)dt=\int_b^0-ds_{rel} \quad .....(2)$$

$$vt_0-\frac{u^2t_0}{v}=b, t=\frac{bv}{v^2-u^2}$$

20. Let y be the height of liquid at some instant.

Then $-\frac{dy}{dt}=$ constant (given)

From equation of continuity,

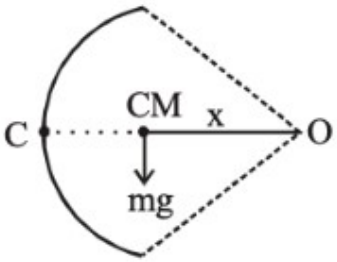
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$\left(\pi x^2\right)\left(-\frac{dy}{dt}\right)=a\sqrt{2gy}$ (a = area of hole)

Here, $\pi\left(-\frac{dy}{dt}\right), a$ and g are constants

Hence, squaring the equation, we get $y=kx^4$.

21.



$I_0=MR^2$

$Mx^2+I_{cm}=MR^2$ .....(i)

& $I_{cm}+M\left(R-x\right)^2=I_C$ .....(ii)

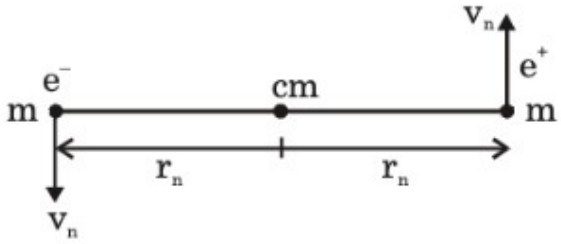
$MR^2-Mx^2+M\left(R-x\right)^2=I_C \qquad \therefore \quad I_C=2MR\left(R-x\right)$

Torque about C.

$Mg\left(R-x\right)=I_c\alpha=\left[2Mr\left(R-x\right)\right]\alpha$

$\alpha=\frac{g}{2R}$

22. Force equation $\frac{ke^2}{\left(2r_n\right)^2}=\frac{mv_n^2}{r_n}$ .....(i)



From quantization of angular momentum

$mv_nr_n+mv_nr_n=\frac{nh}{2\pi} \qquad \Rightarrow 2mv_nr_n=\frac{nh}{2\pi}$ .....(ii)

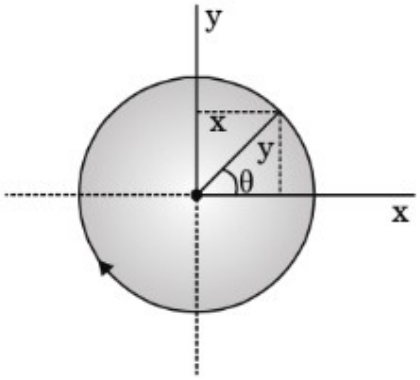
Solving equation (i) & (ii) we get

$r_n=\left(\frac{h^2}{mke^2}\right)n^2; v_n=\left(\frac{ke^2}{2h}\right)\frac{1}{n}$

Total energy $E_n=\frac{\left(PE\right)_n}{2} \qquad E_n=\frac{-ke^2}{2\left(2r_n\right)}$

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23. Consider a current element on the circular ring at an angle θ with the positive x-axis as shown in the figure. The current induced in the ring, $i=\frac{\pi a^2}{R}\left(\frac{dB_z}{dt}\right)$



$\Rightarrow \quad i=\frac{\pi a^2\gamma}{R} \qquad .....(i)$

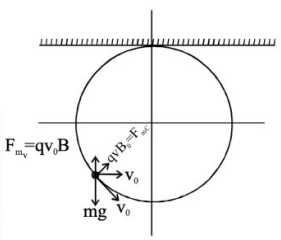
Now, $dF=(iad\theta)(\alpha x\cos\theta)+(iad\theta)(\beta y\sin\theta)$

$\therefore \; F=4i\alpha a^2\int_0^{\pi/2}\cos^2\theta d\theta+4i\beta a^2\int_0^{\pi/2}\sin^2\theta d\theta$

$\Rightarrow \; F=\pi i a^3(\alpha+\beta)$

$\Rightarrow \; F=\frac{\pi^2 a^4(\alpha+\beta)\gamma}{R}$ [using equation (i)]

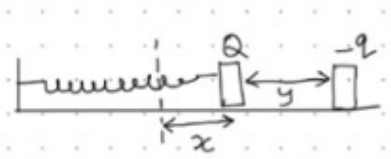
24.



1) $qv_0B=\frac{mv_0^2}{R}$

2) $qv_0B=gm$

25.



$\frac{1}{4\pi\epsilon_0}\frac{Qq}{y^2}=kx \quad ____(i)$

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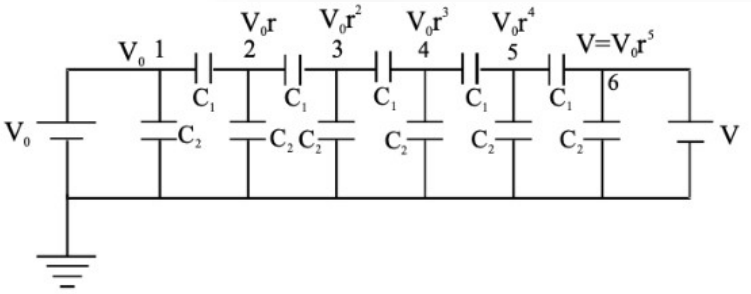
$$\frac{1}{4\pi\in_0}\frac{2Qq}{y^3}dy = kdx$$

For the given condition $dx = -dy$

$$\Rightarrow \text{ extension} = x + y = \frac{3}{2}\left(\frac{Qq}{2\pi\in_0 K}\right)^{1/3}$$

$a = 3, b = 2$

26.



$$C_1(V_0r - V_0) + (V_0r)C_2 + \left(V_0r - V_0r^2\right)C_1 = 0$$

$$r - 1 + \frac{rC_2}{C_1} + r - r^2 = 0$$

$$r - 1 + \frac{4r}{3} + r - r^2 = 0$$

$$\frac{4r}{3} + 2r = r^2 + 1$$

$$3r^2 + 3 - 10r = 0$$

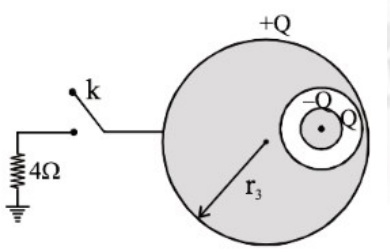
$$3r^2 - 9r - r + 3 = 0$$

$$3r(r-3) - 1(r-3) = 0$$

$$r = 3 \text{ or } \frac{1}{3} \Rightarrow V = V_0r^5$$

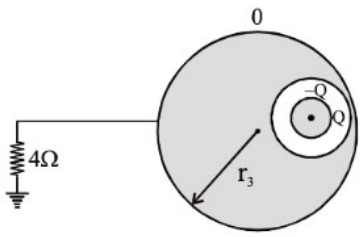
$$V_0 3^5 \text{ or } \frac{V_0}{3^5}$$

27.



Before k close

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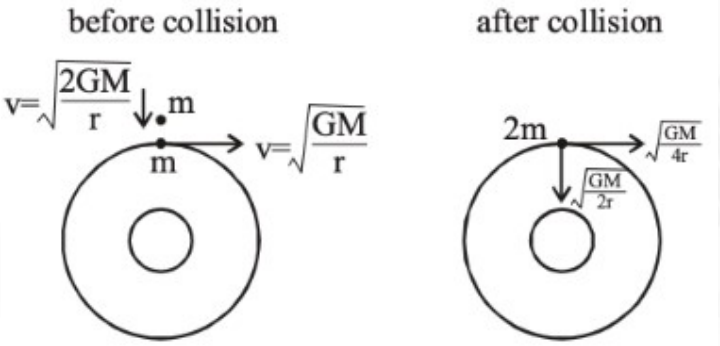
After k close

Heat dissipated = $TE_1 - TE_f$

$= \frac{kQ^2}{2r^3}$ (only difference is self energy of outer surface)

$= \frac{9\times10^9\times4\times10^{-12}}{2\times2} = 9mJ$

28.



By angular momentum $2m\left(\sqrt{\frac{GM}{4r}}\right)r = 2m(V_1)R \Rightarrow V_1 = \sqrt{\frac{GMr}{4}}\frac{1}{R}$

By energy conservation $\frac{1}{2}m\left[\frac{Gm}{4r}+\frac{GM}{2r}\right]-\frac{GMm}{r} = \frac{1}{2}mV_1^2 - \frac{GMm}{R}$

$\Rightarrow -\frac{5GM}{8r} = \frac{1}{2}\left(\frac{GMr}{4R^2}\right)-\frac{GM}{R}$

$\Rightarrow r^2 - 8Rr + 5R^2 = 0 \qquad r = \left(4+\sqrt{11}\right)R$

29. $m\left(2\frac{dr}{dt}\times\frac{d\theta}{dt}+r\frac{dw}{dt}\right) = q\frac{dr}{dt}B$

$\Rightarrow 2mr\frac{dr}{dt}\times\frac{d\theta}{dt}+r\frac{dw^2}{dt} = qrB\frac{dr}{dt} \qquad \Rightarrow m\frac{d}{dt}\left(r^2w\right) = \frac{qrB\,dr}{dt}$

Change in angular momentum of particle $= \int_{r=0}^{r=R} qBrdr = 0$

$\Rightarrow \frac{d\theta}{dt} = 0$ at time when particle leaves the region. So velocity of particle is radially outward.

30. Since chain is loose at the surface, so no tension will generate at the starting point, from where chain comes out from heaps, so using the FBD of falling part of chain

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$p_i = \rho xv \qquad p_f = \rho(x+dx)(v+dv)$

$dp = \rho v + \rho gxdv + \rho gvdx + \rho gdxdv - \rho gxv = \rho gdv + \rho vdx$

$F_x = \frac{dp}{dt} = \rho x\frac{dv}{dt} + \rho v\frac{dx}{dt} \Rightarrow \rho gxdt = \rho(xdv + vdx) = \rho d(vx)$

$gxdt = d(vx)$

Multiplying both side by vx, we have $\Rightarrow gx^2vdt = xvd(xv)$

$\Rightarrow g\int_0^x x^2dx = \int_0^{xv} xvd(xv) \Rightarrow \frac{gx^3}{3} = \frac{x^2v^2}{2} \Rightarrow v = \sqrt{\frac{2gx}{3}}$

$a = \frac{dv}{dt} = \sqrt{\frac{2g}{3}} \times \frac{1}{2\sqrt{x}}\frac{dx}{dt} = \sqrt{\frac{2g}{3}} \times \frac{1}{2\sqrt{x}} \times \sqrt{\frac{2gx}{3}} = \frac{g}{3}$

31. From kinetic energy conservation

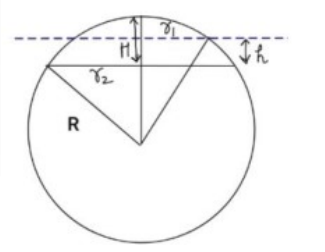
$\frac{1}{2}Mu^2 = \frac{1}{2}mv_{max}^2$

$\Rightarrow v_{max} = \sqrt{\frac{M}{m}}u = 2\ \text{m/s}$

32. If L' be the distance of bigger bead from wall at instant on any collision and v_{rel} be the relative velocity between the beads before collision then it can be shown that

$L(u-0) = L'v_{rel} = L_{min}(v_m - 0) \qquad \Rightarrow L_{min} = L\frac{u}{v_{min}} = \sqrt{\frac{m}{M}}L = 3cm$

33.



$\pi\left(\gamma_2^2 - \gamma_1^2\right) = \pi\gamma_1^2 \quad \gamma_2^2 = 2\gamma_1^2$

$2RH = \gamma_2^2 = 2\gamma_1^2 \quad 2R(H-h) = \gamma_1^2 \quad \Rightarrow h = \frac{H}{2} = 2mm$

34. $\frac{\mu_i}{v} - \frac{\mu_0}{u} = \sum P_r \qquad \frac{4}{3}\left(\frac{1}{30} - \frac{1}{40}\right) = \frac{\frac{4}{3}-1}{R}$

$R = 30cm$

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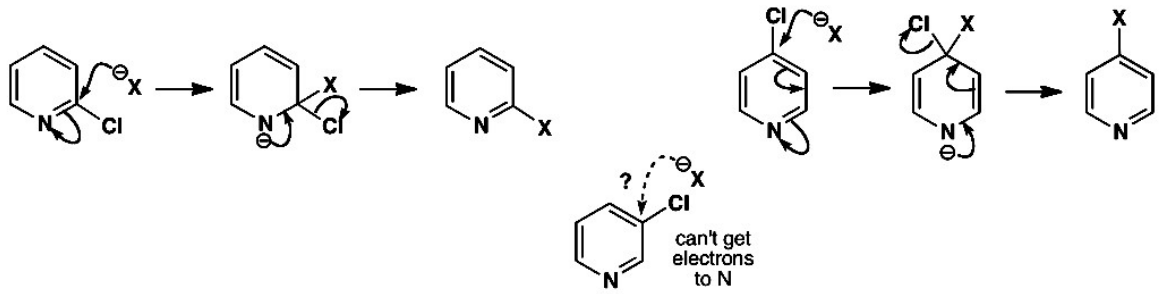
CHEMISTRY

35. Unit cell formula $A_{1/2}BX_{5/2}$ or AB_2X_5

36. The lanthanoid 3+ ions have incomplete $4f$ subshells of electrons. The orbital angular momentum of these electrons contributes to the overall magnetic moment Ga^{3+} has a half-filled f – subshel with a total orbital angular moment of zero. Since there is no orbit angular momentum, it cannot contribute to the moment of the ion and the spin only formula works.

37. All are shows same colour

38.



39. For $M^{n+} + ne^- \rightleftharpoons M$

$$E = E^\circ - \frac{0.06}{n}\log Q$$

For $M^{n+} + ne^- \rightleftharpoons M$ $\quad E = E^\circ - \frac{0.06}{n}\log Q$

$$\Rightarrow E = E^\circ_{M^{n+}/M} - \frac{0.06}{n}\log\frac{1}{\left[M^{x+}\right]}$$

$$\Rightarrow E = E^\circ_{M^{n+}/M} + \frac{0.06}{n}\log\frac{N}{n} \qquad \Rightarrow E = \left(E^\circ_{M^{n+}/M} - \frac{0.06}{n}\log n\right) + \frac{0.06}{n}\log N$$

$n\uparrow$ slope $\downarrow$ Intercept $\uparrow$ if $E^\circ\uparrow$ or $n\downarrow$

The function $\frac{1}{n}\log n$ has its maxima at $n = e$. For integral values, it is maximum at $n = 3$.

40. As the element get more metallic to group, they tend to form positive ions more and oxyanions less. Due to inert pair effect. Tellurium will be oxidized to Te^{+4} next to Te^{+6} by conc. HNO_3.

41. $P: CH_2 = CH - C \equiv CH$ vinyl acetylene
It is formed by reductive coupling of acetylene.
$Q: CH_2Cl - CH = C = CH_2$
It is formed by $1,4$ – addition of HCl to vinyl acetylene
$R: CH_2 = CCl - CH = CH_2$ chloropene
$S: \left(-CH_2 - CCl = CH - CH_2 -\right)_n$ neoprene
It is chain growth polymer formed by free radical addition polymerization.
$T - \{CH_2 - CHCl = CH_2 - CH_2\}_n \leftarrow nCH_2 = CHCl + nCH_2 = CH_2$

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T is copolymer of vinyl chloride and ethene

$$P \xrightarrow[\text{red}]{CuCl+NH_4OH} CH_2=CH-C\equiv C^-Cu^+\downarrow$$

42. Since the $\left[M^{2+}\right]$ varies drastically near equivalence point, we can assume that the reaction has a very high value of equilibrium constant.

$$M^{2+}+Z^{4-} \rightleftharpoons [MZ]^{2-}$$

Case 1:

ini m mol	0.2	0.199	
eq^r	$\sim 10^{-3}$	**?**	~ 0.2

$$M^{2+}+Z^{4-} \rightleftharpoons [MZ]^{2-}$$

Case 2:

ini m mol	0.2	0.201	
eq^r	$\frac{\sim 10^{-3}}{100}$	$\sim 10^{3}$	~ 0.2

$$\therefore K_f = \frac{0.2}{10^{-5}\times 10^{-3}} = 2\times 10^7$$

43. I, II, V & VI

44. n=4, x=26, y=7

45. Inner complex and diamagnetic

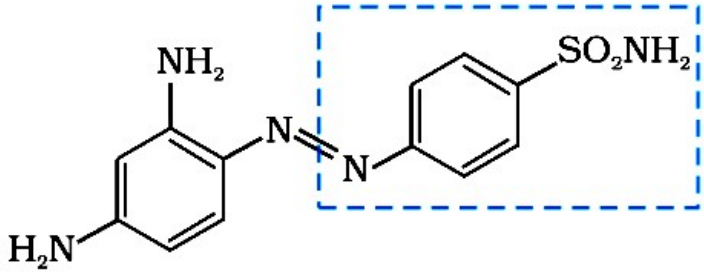
46. $X = 2\,(4^{th}, 5^{th});\ Y = 1$(last one)

47. $II = C_6H_5\overset{+}{N}H_3HSO_4^-$

$III =$ Sulphanilic acid

$IV =$ Sulphanilamide $= X$

$V = m-$Dinitrobenzene



Prontosil

D.U of protosil =11

11+4=15

48, 49.

Rev Ad compression in B

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

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$$\Rightarrow T_2 = 300\left(\frac{12\times A}{1.5\times A}\right)^{2/3}$$

$$\Rightarrow T_2 = 300\times 4 = 1200K.$$

$$n_B = \frac{PV}{RT} = \frac{25\times 0.8}{\frac{1}{12}\times 300} = 0.8\,\text{moles}$$

Now due to heat transfer between A and B, let final temperature be T

$$\therefore n_A C_{V_A}\left(T_f - T_A\right) = n_B C_{V_B}\left(T_B - T_f\right)$$

$$\Rightarrow 1\times\left(T_f - 300\right) = 0.8\left(1200 - T_f\right) \qquad \Rightarrow T_f = 700K.$$

Chamber B:

$\Delta S = 0$ during rev ad compression

During isochoric cooling $\Delta S = nCv_m \ell n\left(\frac{T_2}{T_1}\right) = 0.8\times\frac{3R}{2}\ell n\left(\frac{700}{1200}\right)$

$$= 0.8\times 3\times \ell n7 - 2\ell n2 - \ell n3$$

$$= 0.8\times 3\times\left(2 - 1.4 - 1.1\right) \qquad 2.5 - 2$$

$$= 0.5\times 0.8\times 3 = 1.2\,\text{cal}\,K^{-1}$$

Chamber A:

$$n_{x_{\max}} = n_{y_{\max}} = 0.04$$

$$P_{x_{\max}} = P_{y_{\max}} = \frac{nRT}{V} = \frac{0.04\times\frac{1}{12}\times 700}{1} = \frac{7}{3}\,\text{bar}$$

$Q_p < K_p \Rightarrow$ Complete dissociation takes place

$$p_{He} = \frac{700}{300}\times 25 = \frac{175}{3}\,\text{bar}$$

$$P_{\text{total}} = 63\,\text{bar}$$

50. Chain initiation: $Cl - Cl \longrightarrow 2Cl^{\bullet}; \Delta H = E_a = 84\,\text{K.Cal/mol}$

Chain progation: $CH_3 - CH_2 - H + Cl^{\bullet} \xrightarrow{RDS} CH_3 - \overset{\bullet}{C}H_2 + HCl;$

$\Delta H = +101 - 103 = -2$ K.Cal/mol

51. Disproportionation:

$$\overset{\bullet}{C}H_2 - CH_2 - H + \overset{\bullet}{C}H_2 - CH_3 \longrightarrow CH_2 = CH_2 + CH_3 - CH_3;$$

$$\Delta H = 101 - 101 - \left(146 - 83\right) = -63\,\text{K.Cal}$$

Activation anergy for the reactions between the free radicals is zero.

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